

Exercise 01

① Retrieve all columns from the employees table.

```
SELECT *
FROM employees;
```

Table employees

ID	first-name	Last-name	Department	Salary	Hire-date	City
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-01-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Maidoo	IT	72 000	2017-09-30	Johannesburg

* We use this method to select all the data on every column for every row

② Find all unique departments

```
SELECT Distinct department
FROM employees;
```

Department
IT
HR
Finance
Marketing

③ Retrieve first & last names ordered by salary descending

```
SELECT first_name, last_name
FROM employees
Order by Salary Desc;
```

first_name	Last_name
Eva	Maidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

* Order by Salary Desc with from highest to lowest

④ Retrieve the top 3 highest-paid employees

```
SELECT id, first_name, last_name, Salary
FROM employees
Order by Salary Desc
Limit 3;
```

ID	first_name	Last_name	Salary
5	Eva	Maidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

* Order by Salary Desc with a limit 3.

Filtering with WHERE

⑤ Find employees in the IT Department.

```
SELECT id, first_name, last_name, Department
FROM Employees
WHERE department = 'IT';
```

ID	first_name	last_name	Department
1	Alice	Green	IT
5	Eva	Paideo	IT

* The WHERE statement keeps only rows employees fall under IT Department

⑥ Find employees in finance with Salary > 60000

```
SELECT id, first_name, last_name, department, Salary
FROM Employees
WHERE department = 'finance'
AND Salary > 60000;
```

ID	first_name	last_name	Department	Salary
3	Cathy	Zuber	Finance	65000

* Shows only employees who work in finance and their salary is more than 60000

⑦ Find employees in HR OR Marketing

```
SELECT id, first_name, last_name, Department
FROM employees
WHERE department = HR
OR department = Marketing;
```

id	first_name	last_name	Department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

* It keeps rows if the condition is true

⑧ Find employees not in IT

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department <> 'IT';
```

id	first_name	last_name	Department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

* Looking for data or employees that are not in IT department

⑨ Find employees in IT, HR or finance using IN

```
SELECT id, first_name, last_name, department
FROM employees
Where department IN IT, HR, Finance;
```

Id	first_name	Last_name	Department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Mandoo	IT

⑩ Find employees in IT with Salary > 65000 and City Johannesburg

```
SELECT id, first_name, last_name, department, Salary, City
FROM employees
WHERE department = IT
AND Salary > 65000
AND City = Johannesburg
```